1.3 - Finding Limits Analytically

Goals: Evaluate limits using limit laws, algebra, and the Squeeze Theorem.
Before Class Videos: 1.3.1–1.3.3

Definition:
If $f(x)$ gets arbitrarily close to L for all x sufficiently close to a, then

$$\lim_{x \to a} f(x) = L.$$

For many "nice" functions, evaluating a limit just means plugging in the number. We'll investigate what functions are "nice" in section 1.5 on Continuity. For example:

Example 1:

- $\lim_{x \to 2} x^3 =$
- $\lim_{x \to \pi} \sin(x) =$
- $\lim_{x \to e} \ln(x) =$
- $\lim_{x \to 4} \sqrt{x} =$

Theorem:

If L, M, c, and k are real numbers and

$$\lim_{x \to c} f(x) = L \qquad \text{and} \qquad \lim_{x \to c} g(x) = M,$$

then

1. Sum Rule: $\lim_{x \to c}[f(x) + g(x)] = L + M$
2. Difference Rule: $\lim_{x \to c}[f(x) - g(x)] = L - M$
3. Constant Multiple Rule: $\lim_{x \to c}[kf(x)] = kL$
4. Product Rule: $\lim_{x \to c}[f(x)g(x)] = LM$
5. Quotient Rule: $\lim_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \quad M \neq 0$
6. Power Rule: $\lim_{x \to c}[f(x)]^n = L^n, \quad n$ a positive integer
7. Root Rule: $\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n$ a positive integer

If n is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing c.

Example 2: Evaluate

$$\lim_{x\to 2} \frac{x^2 - 2x}{x + 2}.$$

Example 3: Come up with functions f and g such that

$$\lim_{x\to 3} f(x)g(x) \neq \left[\lim_{x\to 3} f(x)\right] \left[\lim_{x\to 3} g(x)\right].$$

Four Algebraic Tricks

Example 4:

$$\lim_{x\to 2} \frac{x^2 - 2x}{x - 2}.$$

Example 5:

$$\lim_{x\to 0} \frac{(x-2)^2 - 4}{x}.$$

Example 6:

$$\lim_{x \to 9} \frac{\sqrt{x} - 3}{x - 9}.$$

Example 7:

$$\lim_{x \to 2} \frac{\frac{1}{2} - \frac{1}{x}}{x - 2}.$$

Example 8: Evaluate

$$\lim_{x \to 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}.$$

Theorem:

Squeeze Theorem. If $f(x) \leq g(x) \leq h(x)$ for all x in an interval containing a (except possibly at a), and

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L,$$

then

$$\lim_{x\to a} g(x) = L.$$

Example 9: Evaluate

$$\lim_{x\to 0} x^2 \sin(1/x^2).$$

Example 10: Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all x. For what value of a can you determine $\lim_{x\to a} g(x)$? What is the limit?

Suggested after class problems from APEX: Section 1.3 problems 8,11,29,36